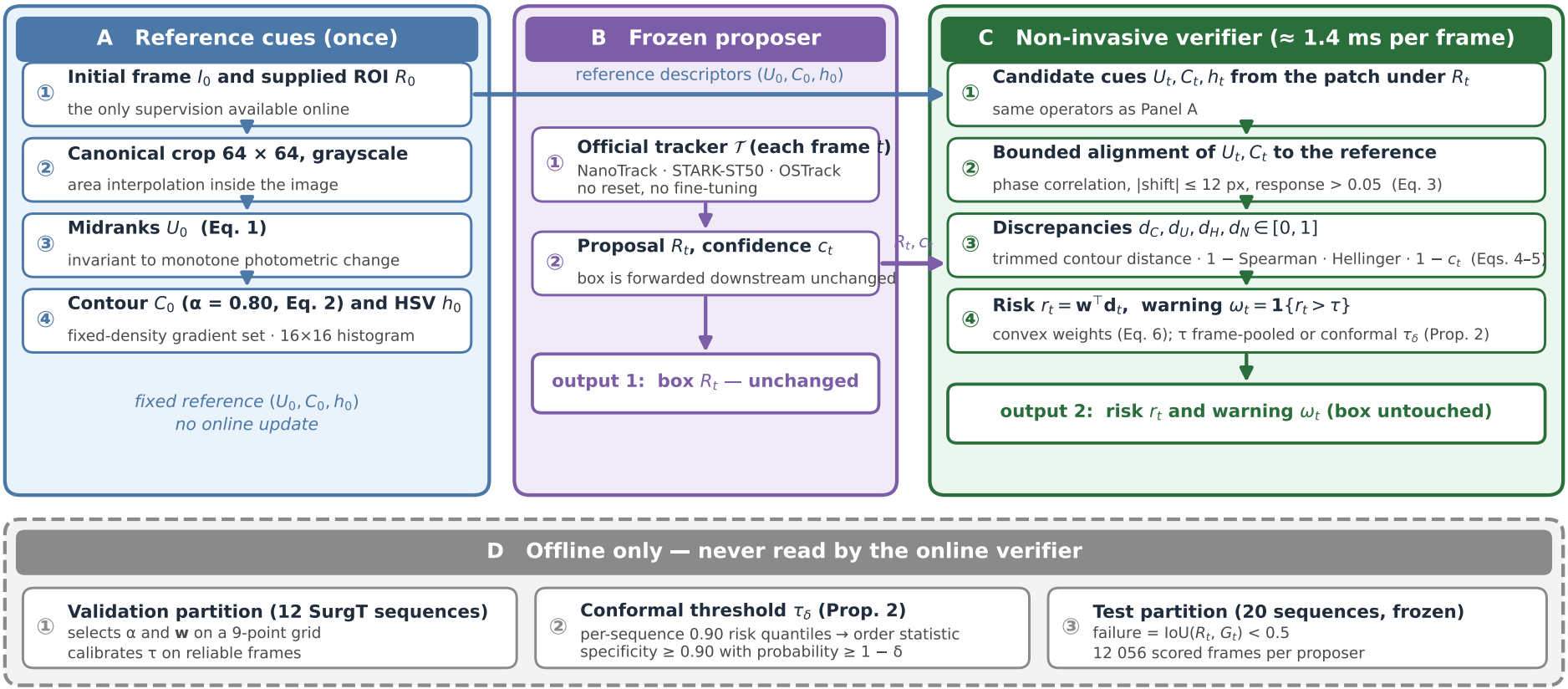

Ordinal-gradient appearance verification around a frozen region-level tracker
online path A → C and B → C at every frame · D is used only offline · the tracker box is never modified
A Reference cues (once)
① Initial frame I_0 and supplied ROI R_0
the only supervision available online
② Canonical crop 64 × 64, grayscale
area interpolation inside the image
③ Midranks U_0 (Eq. 1)
invariant to monotone photometric change
④ Contour C_0 (α = 0.80, Eq. 2) and HSV h_0
fixed-density gradient set · 16×16 histogram
fixed reference (U_0, C_0, h_0)
no online update
B Frozen proposer
reference descriptors (U_0, C_0, h_0)
① Official tracker $\mathcal{T}$ (each frame t)
NanoTrack · STARK-ST50 · OSTrack
no reset, no fine-tuning
② Proposal R_t, confidence c_t
box is forwarded downstream unchanged
output 1: box R_t — unchanged
R_t, c_t
C Non-invasive verifier (≈ 1.4 ms per frame)
① Candidate cues U_t, C_t, h_t from the patch under R_t
same operators as Panel A
② Bounded alignment of U_t, C_t to the reference
phase correlation, |shift| ≤ 12 px, response > 0.05 (Eq. 3)
③ Discrepancies $d_C, d_U, d_H, d_N \in [0, 1]$
trimmed contour distance · 1 − Spearman · Hellinger · 1 − c_t (Eqs. 4–5)
④ Risk $r_t = \mathbf{w}^\top \mathbf{d}_t$, warning $\omega_t = \mathbf{1}\{r_t > \tau\}$
convex weights (Eq. 6); τ frame-pooled or conformal τ_δ (Prop. 2)
output 2: risk r_t and warning ω_t (box untouched)
D Offline only — never read by the online verifier
① Validation partition (12 SurgT sequences)
selects α and w on a 9-point grid
calibrates τ on reliable frames
② Conformal threshold τ_δ (Prop. 2)
per-sequence 0.90 risk quantiles → order statistic
specificity ≥ 0.90 with probability ≥ 1 − δ
③ Test partition (20 sequences, frozen)
failure = IoU(R_t, G_t) < 0.5
12 056 scored frames per proposer